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Synthesis and characterization of novel p-type chemically cross-linked ionogels with high ionic seebeck coefficient for low-grade heat harvesting

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ABSTRACT

The low value of the Seebeck coefficient of the order of few μV/K in inorganic conductors, semiconductors and conducting polymers has inspired the researchers to explore alternative thermoelectric materials. In this work, a novel class of ionogels (IGs) has been synthesized with the aim of developing high-performance thermoelectric materials. Chemically cross-linked ionogels were prepared by the immobilization of ionic liquid, 1-butyl-3-methyl imidazolium tetrafluoroborate ($BMIMBF_4$) in polyethylene glycol dimethacrylate (PEGDMA) matrix using azobisisobutyronitrile (AIBN) as the free radical initiator. The concentration of $BMIMBF_4$ in IGs was varied from 60 to 90 wt (wt. %). The impact of variation of ionic liquid content on the thermoelectric properties of ionogels was analyzed by measuring their thermoelectric properties. Thermal stability, glass transition temperature (T_g), Surface morphology, microstructure, the crystallinity of IGs and nature of chemical interaction between the ionic liquid and polymer matrix were observed by performing TGA, DSC, FESEM, FETEM, XRD, and FTIR respectively. The ionic conductivities of neat $BMIMBF_4$ and IGs were determined by electrochemical spectroscopy using SI 1260 Impedance/Gain-Phase Analyzer. It is worth noting that ionic conductivity (49.41 mS/cm) of IG with 90 wt % of $BMIMBF_4$ is 10 times higher than the ionic conductivity of neat $BMIMBF_4$ (4.5 mS/cm). The origin of this promising achievement (very high conductivity) lies in the “breathing polymer chain model”. The breathing in and out of polymer chains dissociates the ion aggregates resulting in a significant increase in ionic conductivity of IGs. A remarkably higher value of the Seebeck coefficient (2.35 mV/K) was achieved for IG with 60 wt % of $BMIMBF_4$ and defined as ionic Seebeck coefficient because of its origin from the diffusion of ions of the ionic liquid. Due to the positive value of ionic Seebeck coefficient (in an analogy to positive Seebeck coefficient of p-type semiconductors) we termed the ionogels as p-type chemically crosslinked ionogels. A decrease in glass transition temperature of IGs with an increase in ionic liquid content was observed from DSC curves corresponding to an increase in mobility of cations and anions. TGA analysis showed that the synthesized IGs were highly thermally stable up to 390 °C. FESEM and FETEM images revealed that ionic liquid is well confined in the PEGDMA scaffold. The results indicate that ionogels may serve as promising candidates for future thermoelectric applications and also opens the perspective of engineering ionogels with high Seebeck coefficients and electrical conductivities.

1. Introduction

Thermoelectricity offers the possibility to convert low-grade waste heat directly into useable energy in order to fulfill the

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global energy demand [1–3]. For the last 60 years, researchers have investigated various inorganic conductors, semiconductors, conducting polymers and ionic liquids (ILs) for thermoelectric applications. The use of thermoelectric modules is increasing day by day because of their small size, easy to install, no moving parts, light weight, low energy cost and easy to use without having experts [4–6]. The performance of thermoelectric materials can be evaluated by the dimensionless figure of merit, represented by ZT given in the following Equation (1):

$$ZT = \left(\alpha^2.\sigma.T\right)\Big/K \tag{1}$$

where α is Seebeck coefficient, σ is electrical conductivity and K is thermal conductivity and T is the given temperature [7]. The main principle of the traditional thermoelectric modules is based on the Seebeck effect which was discovered by German Physicist Thomas Johann Seebeck in 1821. Both inorganic thermoelectric materials and conducting polymers are characterized by a relatively low Seebeck coefficient of the order of μV/K [8–10]. In the aforementioned materials, it is also very challenging to overcome the interaction between electrical and thermal conductivity. Newer, Liquid state thermoelectrochemical technologies emerged as promising substitutes and less expensive routes to harvest the thermal energy into electrical energy without polluting the environment. These devices utilize redox couples with an electrolyte such as ionic liquid. Ionic liquids have many interesting properties, for example, negligible vapor pressure, superior thermal stability, high ionic conductivity and low thermal conductivity [11–14].

Theodore J. Abraham et al. have demonstrated a high Seebeck coefficient from 1.5 to 2.2 mV/K using ILs and redox couple [15]. Recently, Duan et al. achieved very high Seebeck coefficient of 4.2 mV/K from aqueous thermogalvanic cells [16]. However, leakage problems due to their liquid state prevented the device fabrication for potential thermoelectric applications. This problem can be avoided by immobilizing the ILs into the polymer matrix. In this way, the conductive nature of the ionic liquids is preserved as well as their leakage blocked through the molecular architecture of the polymer matrix [17]. Ionogels (IGs) are the class of materials whereby ILs are confined in the polymer matrix by various methods. The simplest way to prepare cross-linked ionogels is by soaking the polymer membrane in ionic liquids although it does not permit the control of the exact amount of IL absorbed. Another drawback of physically cross-linked ionogels is that the networks of non-covalent associations such as hydrogen bonding and host-guest associations are easily broken and show weak mechanical and thermal stability [18]. In situ thermally initiated radical polymerization of monomers in the presence of IL as a solvent medium is the best approach to prepare the chemically cross-linked ionogels. In this way, ILs are well confined and homogeneously distributed in the polymer matrix and cross-linking improves the mechanical and thermal stability of IGs [19]. The ionogels have many industrial applications such as dye-sensitized solar cells (DSSCs) for solar energy conversion, electrochemical double layer capacitors (EDLCs), electrochemiluminescent materials, and actuators [20–22]. In our previous work thermally initiated free radical polymerization of polyethylene glycol dimethacrylate was used to synthesize crosslinked thermoelectric hydro-ionogel (X-TEHIG) from triethyl octyl ammonium bromide ([N_{2228}] Br) as an ionic liquid. Unlike conventional thermoelectric materials which have electronic Seebeck coefficient (because of motion of electrons) the origin of Seebeck coefficient in ionogels is a motion of cations and anions, so we termed it as ionic Seebeck coefficient. Significantly higher values of ionic Seebeck coefficient (1.38 mV/K) of X-TEHIG was achieved. But the thermal stability of ionic liquid bromide ([N_{2228}] Br) was not too high and consequently, the thermal stability of X-TEHIG was also affected by the IL. Ionic conductivity of an ionic liquid is mainly attributed to the motion of its cations because it has a higher diffusion coefficient than that of the anion. Ionic liquids based on imidazolium cations demonstrated high ionic conductivity values but when both high ionic conductivity and high thermal stability are desirable then imidazolium cation with boron tetra fluoroborate anion is the best choice. Because of the above-mentioned reason we used 1-butyl-3-methyl imidazolium tetra fluoroborate ($BMIMBF_4$) in this study. The aforementioned in situ radical polymerization methodology is adopted to synthesize PEGDMA supported ionogels of $BMIMBF_4$ and concentration of $BMIMBF_4$ is varied from 60 to 90 wt %. The effect of variations in $BMIMBF_4$ content on thermoelectric and thermophysical properties of ionogels have been investigated and analyzed in recent work. It might be highlighted that thorough research on relevant literature yielded no report on the thermoelectric characterization of chemically cross-linked ionogels. In order to observe the microstructure and crystallinity of IGs, FESEM, FETEM and XRD analysis were carried out respectively. Thermal stability and pathways of thermal decomposition have been examined by thermal analysis methods using DSC and TGA.

2. Methodology

2.1. Materials

All the materials and chemicals were used as received without any further processing. 1-butyl-3-methyl imidazolium tetrafluoroborate ($BMIMBF_4$) was received from Merck, Germany. Additionally, polyethylene glycol 200 Dimethacrylate was purchased from Geo Specialty Chemicals, the USA and the CR2032 stainless steel coin cell battery casings with grade 316SS were obtained from GELON Lib Co limited, China.

2.2. Synthesis of ionogels

Ionogels were synthesized by thermally initiated free radical polymerization of PEGDMA. In this process, the ionic liquid was immobilized in a polymer matrix using AIBN as a free radical initiator. Firstly, $BMIMBF_4$ (IL) and PEGDMA were mixed homogeneously by stirring at room temperature and then AIBN powder was poured into the mixture. The precursor was stirred and heated at 60 °C for 10 min to fully dissolve AIBN. Next, the mixture was heated at 90 °C for 15 min and ionogels were formed. Fig. 1 (a) indicates the successful synthesis of IGs as it is confirmed from the

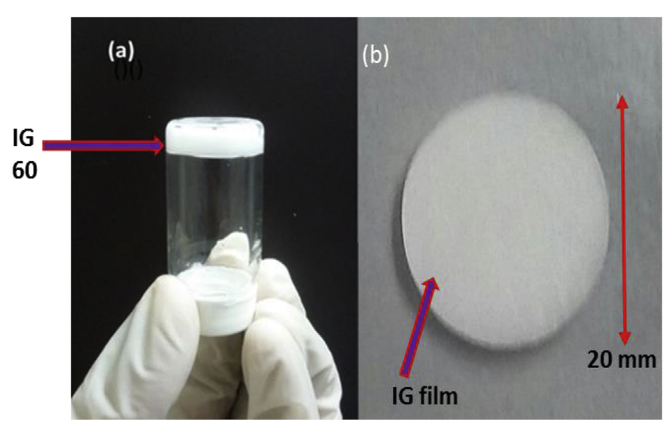

Fig. 1. (a) Ionogel 60 showing inversion test (b) Ionogel film with a diameter of 20 mm and thickness of 2 mm delicately cut into round disks and fabricated inside coin cell for electrical conductivity, Seebeck coefficient, and Power output measurements.

inversion test and Fig. 1 (b) shows the dimensions of synthesized ionogels used in the fabrication of coin cells. The contents of $BMIMBF_4$ were varied by 60, 70, 80, and 90 wt % to prepare four types of ionogels. From now onwards, we will write IG 60, IG 70, IG 80, and IG 90, having 60, 70, 80 and 90 wt % of $BMIMBF_4$ respectively. Fig. 2 (a) shows the schematics for the synthesis of IGs. In the first step PEGDMA, $BMIMBF_4$ and AIBN are mixed homogeneously. Then AIBN generates free radicals and in the final step, ionogel is formed. Fig. 2(b) designates the chemical structures of ionic liquid, 1-butyl-3methyl imidazolium tetra fluoroborate ($BMIMBF_4$), polymer, polyethylene glycol dimethacrylate (PEGDMA) and free radical initiator azobisisobutyronitrile (AIBN). In Fig. 2(C) part (a) overall reaction for the synthesis of ionogels is shown. The detailed mechanism is shown in Fig. 2(C) part (b) which describes the initiation propagation and termination steps involved in the synthesis of chemically cross-linked ionogels by free radical polymerization.

2.3. Characterizations

Impedance spectroscopy measurements were carried out as a function of temperature using a Solartron SI 1260 Impedance/Gain-Phase Analyzer with the frequency range from 0.1 to 10^6 Hz with AC amplitude of 10 mV. The ionic conductivity and Seebeck coefficient measurements were performed with the fabrication of a coin cell using CR2032 (with a diameter of 20 mm and a thickness of 3.2 mm) battery casings. The thickness of IGs sandwiched between stainless steel electrodes was 2 mm measured by a digital Vernier Caliper. Furthermore, in order to find the temperature dependence of impedance, the coin cells were mounted on a sample holder and heated in an Espec SU-241 temperature chamber with the temperature fluctuation of ± 0.3 °C. Once the set temperature is reached in the chamber, the measurement was taken after 10 min in order to ensure the temperature uniformity of the sample. The Seebeck coefficient was investigated by creating the temperature gradient between two electrodes of a coin cell by using a Peltier cooler and a ceramic heater. The KEITHLEY 2231A-30-03 Triple Channel DC power supply was used as a power source. Before recording any value of voltage, ΔT was maintained for 2 h with a maximum fluctuation of ± 0.5 °C for hot side or cold side. For each measurement after increasing the power supply voltage, it was paused for an initial 10 min to equilibrate the temperature. For the next 2 min, for each measurement five values of open circuit voltage (V_{oc}) were recorded by an Agilent 34461A$6_{1/2}$ digital Multi-meter. The thermocouple wires were attached to hot and cold sides of a coin cell in order to note the temperature differences. The potential difference between the two electrodes increased linearly with temperature. Temperature gradients of $\Delta T = 10$ K, 20 K and 30 K were maintained across the two sides of coin cell and corresponding values of open circuit voltage (V_{oc}) were recorded and the graph was plotted between ΔT and (V_{oc}). The schematics of experimental set up for measurement of Seebeck coefficient is shown in Fig. S1. The value of the Seebeck coefficient was calculated from the slope of the graph between V_{oc} and ΔT. Additionally, for the measurement of power output, the same procedure was followed as described by Hasan et al. [23]. The Transient Hot Bridge (THB) 500 instrument from Linseis (Germany) with a power heater of 20 mW and measuring the current of 5 mA was employed to measure the thermal conductivities of the IL and ionogels. The thermal conductivity measurement range of THB 500 is from 0.01 up to 500 W/(m.K) with an accuracy better than 5%. The recently developed Quasi-Steady-State (QSS) method is the principle of THB 500 for the measurement of thermal conductivity. The Hotpoint sensor with the small size of 4.5 mm is a development of the QSS sensor and allows measurements of a big range of applications, with small sample sizes. Due to

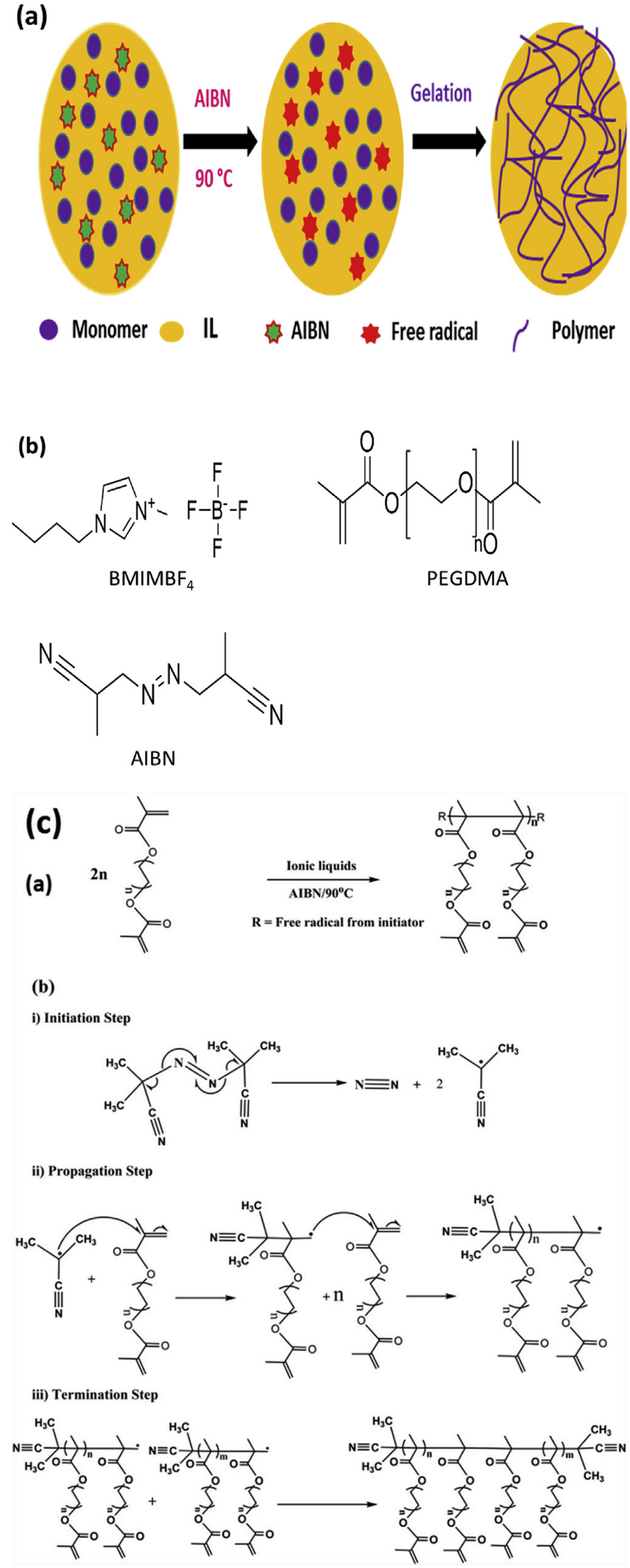

Fig. 2. (a). Schematics for the synthesis of ionogels by thermally initiated free radical polymerization of PEGDMA monomers in presence of $BMIMBF_4$ (IL) and AIBN as a radical initiator to prepare IGs. 2(b) Chemical structures of $BMIMBF_4$ (IL), PEGDMA and AIBN. 2(c) part (a) chemical reaction for the synthesis of ionogels and 2(c) part (b) steps involved in the synthesis of chemically cross-linked ionogels.

its small size, side effects can be ignored.

Thermogravimetric analysis of the IL, PEGDMA, and ionogels was carried out using PerkinElmer TGA 4000. The alumina crucibles (180 μL) which can withstand the up to a temperature of 1750 °C under an ultra-pure nitrogen flow rate of 19.8 ml/min and a heating rate of 10 °C/min from 30 to 500 °C were used for thermogravimetric analysis. The obtained data was analyzed using Pyris Software. A differential scanning calorimetry (Setaram DSC 131 Evo) in the temperature range between −50 and 190 °C with a scan rate of 10 °C/min was used to measure the glass transition temperature of the samples. The samples were tightly sealed in a regular aluminum pan of 30 μL under a nitrogen atmosphere with a flow rate of 20 ml/min. Infrared spectra were recorded using the PerkinElmer Spectrum Two FTIR spectrometer from 350 to 4000 cm^{-1} running with 16 scans and resolution of 4 cm^{-1}. In addition, the base line and ATR corrections have been done to all samples. In order to investigate the electrochemical behavior of IGs, cyclic voltammetry was performed with a potential range of −1.5–1.5 V at a scan rate of 75 mVs^{-1}. The ionogel crystal structure was determined by X-ray diffraction using benchtop powder X-ray diffraction (Rigaku Miniflex) with Cu Kα radiation (λ = 1.5406 Å) operating at 40 kV and 15 mA. XRD was carried out in the range of 3–90° with a scan rate of 10°/min and a step width of 0.02°. The degree of crystallinity of ionogels was calculated using Origin 2019b software. The surface morphology of the ionogels was investigated by Field emission scanning electron microscope (FESEM) using the HITACHI SU8010. The ionogel samples were platinum coated by using the Quorum (Q150R S) rotary pumped coater with a sputtering current of 30 mA before scanning in FESEM. Elementary analysis mapping of FESEM images was performed using X-max Silicon drift X-ray detector from Horiba. The microstructure of the IGs was observed by field emission transmission electron microscopy (FETEM, JEM-2100F, JEOL Inc., Japan) at an accelerating voltage of 200 kV. The samples for FETEM were prepared by mixing the IGs with water and methanol followed by heating at 90 °C for 5 h to achieve a homogeneous mixture. Then the diluted IGs were dropped on a carbon-coated copper grid and air dried for 24 h.

3. Results and discussion

3.1. Attenuated total reflectance-fourier transforms infrared spectroscopy (ATR-FTIR)

ATR-FTIR spectroscopy was performed to observe the possible ion-polymer interactions and conformational changes in the PEGDMA polymer matrix. FTIR spectra of $BMIMBF_4$ (neat ionic liquid), PEGDMA (pure polymer) and four types of ionogels are shown in Fig. 3(a). The symmetric (C−H) stretching vibration of the imidazolium produced peaks at 3161 cm^{-1} and 3120 cm^{-1} in the spectrum of $BMIMBF_4$. The peaks at 521, 623, 754, 848 and 1035 cm^{-1} were attributed to B−F stretch, imidazole ring vibration, out of plane imidazole C−H bending, in-plane imidazole C−H bending and B−F stretch in BF_4 respectively. The peaks in range from 2800 to 3000 cm^{-1} are ascribed to butyl chain attached to imidazolium ring. The stretching vibration of C=N and bending vibration of C−H are observed at 1169 and 1574 cm^{-1} respectively. FTIR spectrum of $BMIMBF_4$ was in good agreement with the literature [24–27]. Any interaction between $BMIMBF_4$ and PEGDMA will lead to changes in the frequency of absorption bands. In the spectrum of PEGDMA, the peak at 2872 cm^{-1} is for symmetric C−H stretch of a methyl group and the characteristic intense peak at 1715 cm^{-1} represents carbonyl group (C=O) of methyl acrylate group. The peaks at 941 and 1636 cm^{-1} were assigned to C=C stretch of methacrylate group in PEGDMA. The band connected with C−O group of esters was observed at 1165 cm^{-1} [28,29]. All

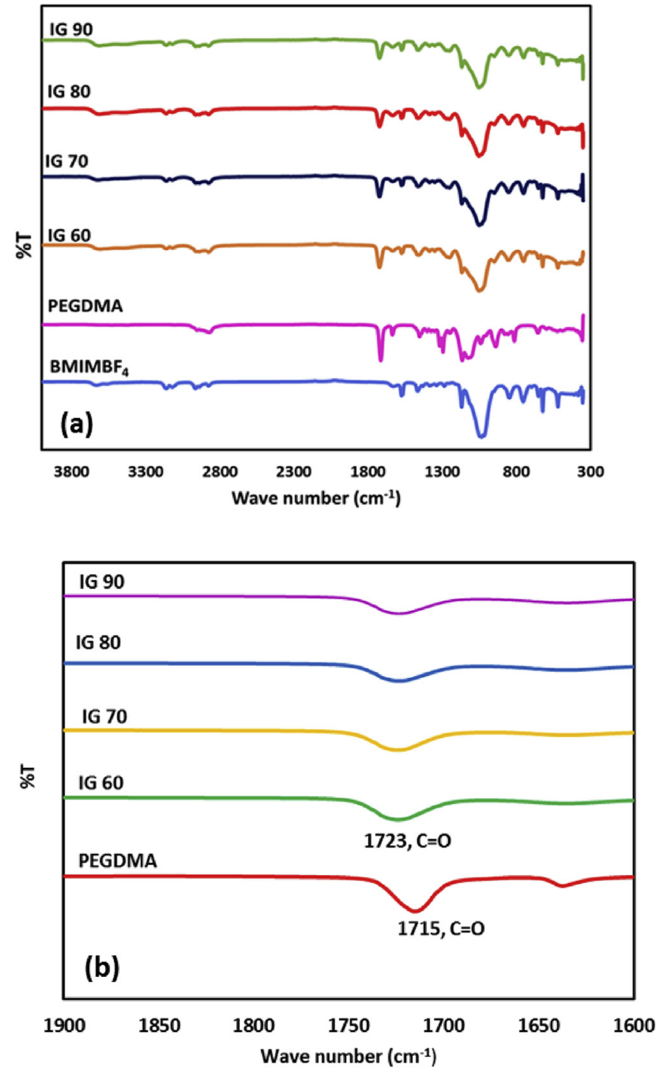

Fig. 3. (a). FTIR Spectra of $BMIMBF_4$, PEGDMA, and four types of ionogels. FTIR spectra were recorded using the PerkinElmer Spectrum Two FTIR spectrometer from 350 to 4000 cm^{-1} running with 16 scans and resolution of 4 cm^{-1}. (b) Displays the shifting of C=O peak from 1715 cm^{-1} to 1723 cm^{-1}.

spectra of IGs were dominated by either the peaks from PEGDMA or the peaks from $BMIMBF_4$. The C=O peak in PEGDMA monomer was observed at 1716 cm^{-1} and it was shifted to 1723 cm^{-1} in spectrum of PEGDMA ionogels as indicated in Fig. 3(b). The complete shifting of C=O peak from 1715 cm^{-1} to 1723 cm^{-1} without any residual peak at 1715 cm^{-1} shows that no C=O bond is conjugated with C=C bonds all the PEGDMA monomers are used up in the reaction [30]. The bands derived from the $BMIMBF_4$ were slightly shifted to higher or lower wave numbers. But the most noticeable shift was observed in B−F stretch in BF_4 anion of $BMIMBF_4$. It was shifted from 1035 cm^{-1} to 1051 cm^{-1}, 1051 cm^{-1} 1053 cm^{-1} and 1055 cm^{-1} in IG 60, IG 70, IG 80 and IG 90, respectively. This indicates a strong interaction between BF_4 anion and PEGDMA matrix in the form of complexes which may include non-covalent interactions. The peak at 815 cm^{-1} corresponding to C=C in PEGDMA disappeared in spectrum in all the IGs indicating the successful polymerization.

3.2. XRD analysis of ionogels

XRD was performed to evaluate the crystal structure and degree of crystallinity of ionogels. XRD pattern of ionogels revealed their semi-crystalline morphology i.e. a structure mixed with both crystalline and amorphous regions as shown in Fig. 4. The crystalline peaks were observed at $2\theta = 27.23°$, $29.21°$, $35.79°$, $39.21°$,$42.92°$, and $47.21°$ in XRD pattern of all ionogels at fixed positions which indicate that the confined ionic liquid remained in the form of ion pairs. However, the intensity of all the peaks showed a gradual decrease with the increase in the ionic liquid content with the highest intensity for IG 60 and the lowest intensity for IG 90. Wu et al. also observed the fixed peak positions in XRD pattern of ionogels and decrease in peak intensity with increase in IL content [31]. Besides the crystalline peaks, a broad hump is noticed at $2\theta = 20.21°$ in XRD pattern of IG 70 showing the amorphous region of ionogels. With the further addition of ionic liquid into IGs, two broad humps were observed at $2\theta = 10.7°$ and $22.67°$ in IG 80 and IG 90. The reduction in the intensity of crystalline peaks and the introduction of broad peaks implies less crystalline and more amorphous nature of IGs. The degree of crystallinity of ionogel was calculated by using Equation (2).

$$\text{Degree of Crystallnity} = (\textit{area of crystalline peaks})/ \times (\textit{area of all peaks}) \times 100 \tag{2}$$

The degree of crystallinity of IG 60 was 31.26 and with the addition of 70 wt % of IL into the PEGDMA matrix, it decreased to 27.90 for IG 70. The lowest degree of crystallinity equal to 12.67 was observed for IG 90. A decrease in crystallinity was also observed in a study of tough butyl methyl imidazolium chloride (BMIMCl) based ionogel published by Liu et al. [32]. Thus, an increase in the concentration of ionic liquid from 60 to 90 wt % has significantly increased the amorphicity of ionogels by creating disorder in PEGDMA chains and making them more flexible. Aforementioned structural changes in polymer chains promote the ionic transport in ionogels and provide evidence for an increase in ionic conductivity of ionogels.

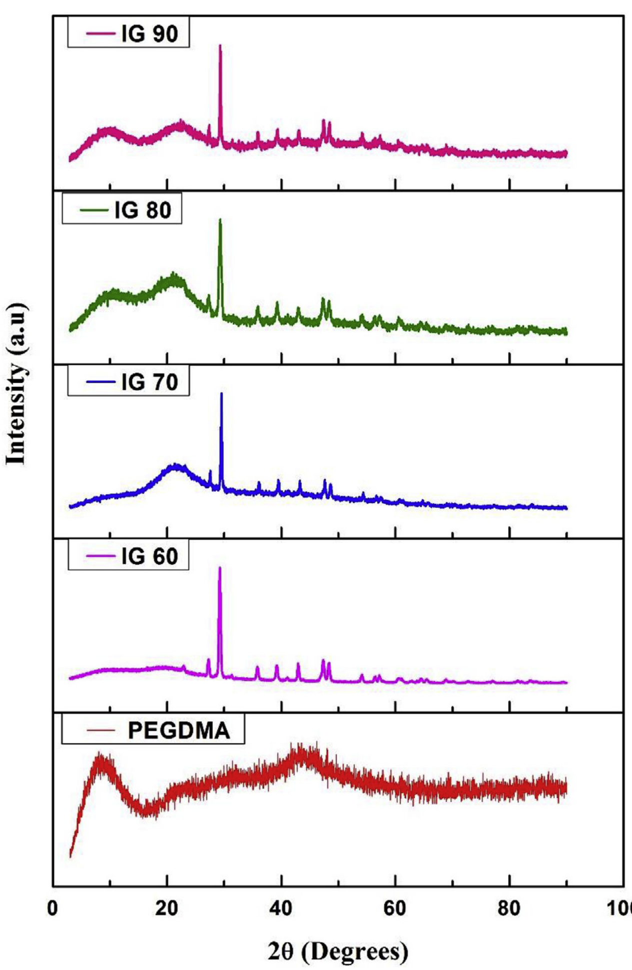

Fig. 4. XRD graphs of PEGDMA and four types of ionogels. XRD was carried out in the range of 3–90° with a scan rate of 10°/min and step width of 0.02°.

3.3. Thermal analysis of ionogels

In the thermal analysis, the response of the samples is measured with respect to a predetermined temperature range. The widely used techniques are the thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) [33]. DSC thermograms of ionogels with different loadings of IL from a heating range of −50 to 190 °C are shown in Fig. 5. The glass transition temperatures (T_g) were determined from the midpoint of heat capacity change in DSC thermograms during the heating of ionogels. From the DSC thermograms, it is clear that glass transition temperatures (T_g) of ionogels decreases with the increase in IL concentration from 33.4 °C for IG 60 to 25 °C in the IG 90. When $BMIMBF_4$ molecules are confined in PEGDMA matrix they interrupt the polymer interactions and create disorder in the original crystalline structure of polymers. Thus, enhancement of $BMIMBF_4$ content in PEGDMA matrix increases disorder and lower the glass transition temperature of ionogels [34,35]. This observation is also consistent with the XRD analysis of IGs discussed in section 3.2 of this paper. As we have described in the interpretation of XRD analysis of ionogels in section 3.6 that the increase in the content of IL has decreased the degree of crystallinity of IGs rendering them amorphous. Therefore, DSC results indicate that the increase in $BMIMBF_4$ content has a great influence on the motional dynamics of cations and anions of the $BMIMBF_4$ present in IGs. The decrease in T_g of IGs also supports the increase in ionic conductivity of IGs by increasing the mobility of cations and anions according to Equation (3).

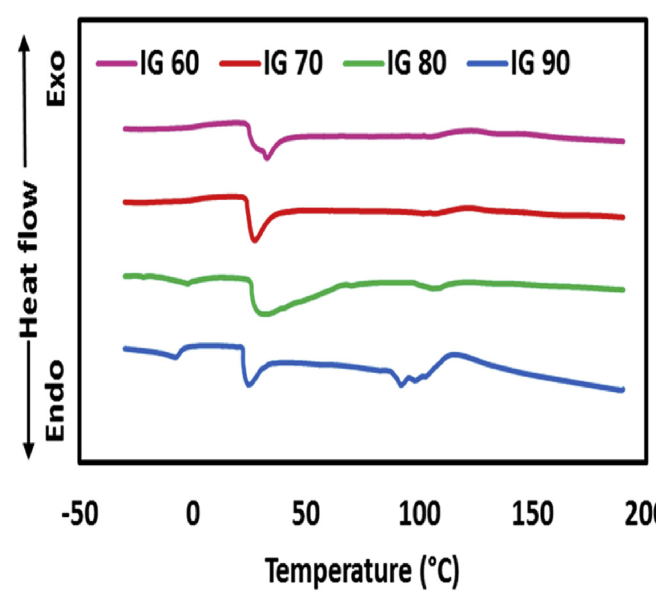

Fig. 5. DSC thermograms of ionogels. A differential scanning calorimetry (Setaram DSC 131 Evo) in the temperature range between −50 and 190 °C with a scan rate of 10 °C/min was used to measure the glass transition temperature of the samples.

$$\sigma = n\, q\, \mu \tag{3}$$

σ (Sm^{-1}) where is ionic conductivity, n is no of charge carriers q is the mount of charge (C), and is μ (m^2/V.s) mobility of charge carriers. The thermal degradation behavior of all samples is shown in Fig. 6 obtained by TGA.

The T_{onset} (value of the temperature where any measurable weight loss is observed) in the TGA curve of $BMIMBF_4$ was observed at 325 °C. Fredlake et al. studied the thermophysical properties of imidazolium-based ionic liquids and reported T_{onset} for $BMIMBF_4$ at 315 °C [36]. The 10 °C difference of our measured value may be attributed to the origin of the ionic liquid or composition of pan used in experiments. The major weight loss for $BMIMBF_4$ occurred at 400 °C and it reached 98% until 490 °C. The similar behavior of thermal stability of $BMIMBF_4$ has been reported by Ammam et al. [37]. The thermal decomposition of $BMIMBF_4$ is attributed to the thermal cleavage of C—N bond in $BMIMBF_4$ [38].

The T_{onset} for PEGDMA curve was observed at 175 °C and rapid weight loss started after 250 °C because of decomposition of crosslinked polymer chains. PEGDMA totally degraded until 450 °C. This trend coincides with that published by Kim et al. [39]. From TGA thermograms, it can be observed that the thermal stability of IGs increased with an increase in $BMIMBF_4$ content. The TGA curve of IG 60 demonstrated a T_{onset} at 271 °C which indicates that thermal stability of ionogel after immobilization of ionic liquid in the PEGDMA matrix is improved as compared with pure PEGDMA. This increase in thermal stability can be explained with the help of FTIR results. As we have mentioned earlier in the discussion of the FTIR portion of this paper that the peak for B—F stretch of BF_4^- was observed at 1035 cm^{-1} in the FTIR spectrum of neat $BMIMBF_4$. In the FTIR spectrum of IG 60, the above-mentioned peak is shifted to higher wave number of 1051 cm^{-1} which indicates that BF_4^- anion makes strong complexes with PEGDMA matrix which are responsible for the improvement in thermal stability of ionogels. Moreover, % transmission of light through B—F stretch in the spectrum of is 65.89% which is reduced to 33.12% for the same band in IG 60. The reduction in % transmission of light corresponds to more absorption and this provides evidence for the presence of a higher number of BF_4^- complexes with a polymer which contribute to the higher thermal stability of ionogels. Another reason for the improved thermal stability of ionogels is crosslinking of polymer chains in the presence of dimethacrylate as cross linker during the free radical polymerization process. Upon further increase of the amount of ionic liquid, the T_{onset} for ionogels increased and the highest thermal stability up to 390 °C was observed for IG 90. Shalu et al. have also reported an increase in thermal stability of ionogels (synthesized from PVDF-HFP and $BMIMBF_4$) with the increase in the amount of $BMIMBF_4$ [40]. The excellent thermal stability of ionogels ensured their applicability in high-temperature thermoelectric applications.

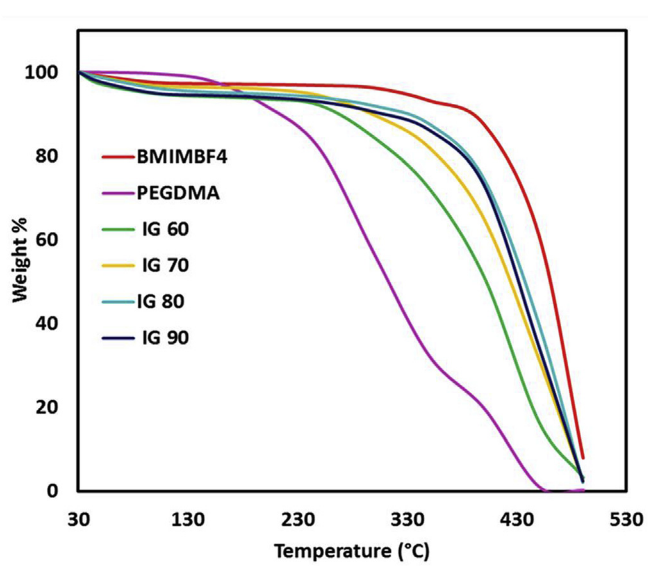

Fig. 6. TGA thermograms of $BMIMBF_4$, PEGDMA, and four types of ionogels. Thermogravimetric analysis of the IL, PEGDMA, and ionogels were carried out using PerkinElmer TGA 4000. The alumina crucibles (180 μL) which can withstand the up to a temperature of 1750 °C under an ultra-pure nitrogen flow rate of 19.8 ml/min and a heating rate of 10 °C/min from 30 to 500 °C were used for thermogravimetric analysis.

3.4. Microstructural studies and EDX elemental analysis of ionogels

The IGs possess a smooth and crack-free surface and IL is encapsulated into PEGDMA scaffold as shown by FESEM images in Fig. 7. The spot analysis and elemental composition of the ionogels at different points were evaluated by EDX as shown in Fig. S2 (a) and (b) respectively. Remarkably, the weight % of all the elements at different points in IGs samples was almost the same. The presence of Boron, Nitrogen, and Fluorine in all parts of the IGs proved that $BMIMBF_4$ is homogeneously distributed and successfully confined in the PEGDMA polymer matrix. The elemental mapping of IGs describing the spatial distribution of various elements which is depicted in Fig. S3. Crosslinking of polymer chains is evident from FETEM images as shown in Fig. 8.

3.5. Ionic conductivity

The ionic conductivity (σ) of neat $BMIMBF_4$ at 30 °C was 4.56 mS/cm which is in very good agreement with the value of 4.5 mS/cm reported by Tokuda et al. at the same temperature [41]. The ionic conductivity of IL increases with temperature as shown in Fig. 9. This increase in σ is attributed to decreases in viscosity of IL with an increase in temperature. Uhl et al. (2014), reported a decrease in viscosity of pure $BMIMBF_4$ from 0.1007 kg/(s. m) at 25 °C to 0.0177 kg/(s. m) at 70 °C [42]. The purpose of the measurement of ionic conductivity (σ) of ionogels was to observe the effect of $BMIMBF_4$ concentration on electrical properties of ionogels at four different loadings of $BMIMBF_4$ from 60 to 90 wt %. Nyquist plots of BMIMBF4 and all IGs are shown in Fig. S4. The value of impedance was calculated by linear fitting of Nyquist's plots using Z View software. Equation (3) was used to compute the ionic conductivity of IL and IGs.

$$\sigma = \frac{L}{A.R_b} \tag{3a}$$

where L (cm), A (cm^2) and R_b (Ω) are the thickness of the ionogel, area of the electrodes and bulk resistance of the IGs respectively.

The σ of IGs has increased with the increase in the concentration of $BMIMBF_4$ with a value of 9.1 mS/cm in IG 60 and 47.75 mS/cm in IG 90 at 30 °C as shown in Fig. S5. The origin of this increase in ionic conductivity is the increase in the number of ions with an increase in the concentration of $BMIMBF_4$. The ionic conductivity of ionic liquid and ionogels at different temperatures is presented in Fig. 3. The ionic conductivity of IL and IGs demonstrated an increasing trend with an increase in temperature which shows the behavior of semiconductors [43]. Susan et al. reported that high self-dissociation of ions, ion transport ability of the constituent IL and decoupling of ion transport from the segmental motion are mainly responsible for high ionic conductivity of the IGs [44]The highest value of σ (76.78 mS/cm) was achieved for IG 90 at a temperature of

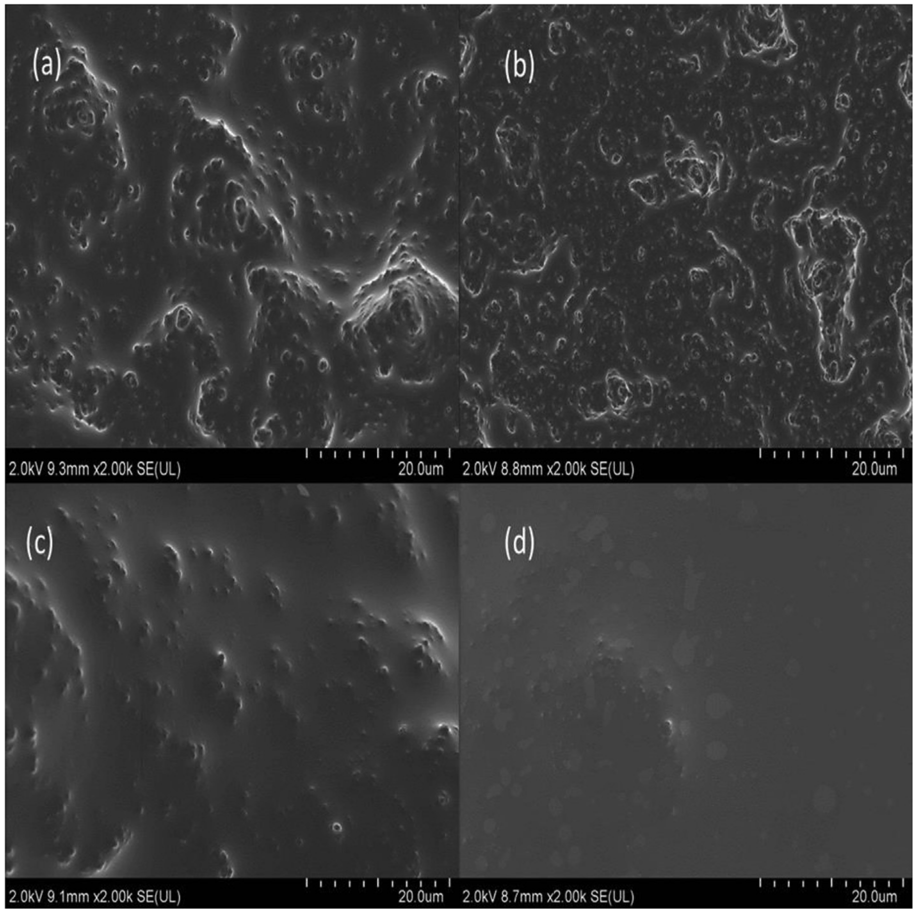

Fig. 7. FESEM images of (a) IG60, (b) IG 70, (c) IG80, and (d) IG 90. The ionogel samples were platinum coated by using the Quorum (Q150R S) rotary pumped coater with a sputtering current of 30 mA before scanning in FESEM (HITACHI SU8010).

90 °C. This reflects a large number of ionic species and lower viscous environment that facilitates the ionic motion. Upon further increase of the temperature above 90 °C, no significant increase in ionic conductivity of IGs was observed. The possible assumption is that with an increase in temperature the dissociation of cations and anions increases, which may decrease the mobility of cations and anions. Consequently, the conductivities of all the IGs are considerably higher than that of pure $BMIMBF_4$. Datta et al. showed higher ionic conductivity of polymer electrolyte as compared with the constituent ILs, 1-ethyl 3-methyl imidazolium chloride (EMIMCl) and 1-butyl-3-methyl imidazolium bromide (BMIMBr) with the low value of ionic Seebeck coefficient (17.92 μV/K). The possible reason for the low value of the Seebeck coefficient is that they have physically dispersed the IL into a polymer matrix in which the IL may not be homogeneously distributed as compared with our chemically crosslinked ionogels. This will be discussed in more detail in the discussion of the Seebeck coefficient. Another reason for the higher conductivity of IGs is the "breathing polymeric chain model". This model states that intrinsic folding and unfolding of polymer chains induce fluctuations inside polymer scaffolds at the microscopic level which increases ionic motion compared to neat ILs. The breathing in and out of polymer chains dissociates the ion aggregates resulting in a remarkable increase in ionic conductivity of IGs [45,46]. The DSC curves of IGs provide experimental evidence for the increment in ionic conductivity with the increase of $BMIMBF_4$ content as glass transition temperature (T_g) of IGs decreases from 33.4 to 25 °C for IG 60 and IG 90 respectively. The decrease in T_g increases the mobility of ions and facilitates ionic transport in ionogels. The decrease in the degree of crystallinity with an increase in the amount of IL also supports the increment in ionic conductivity of IGs.

3.6. Seebeck coefficient

The definition of Seebeck coefficient describes the ability of a material to convert temperature gradient potential difference and is given by equation (4).

$$S_i = V_{thermal}/\Delta T \tag{4}$$

where S_i is the ionic Seebeck coefficient, $V_{thermal}$ is the potential difference produced and ΔT is temperature difference across the electrodes [43]. The potential difference is generated due to the transport of mobile charge carriers by heat flux. The value of the ionic Seebeck coefficient of pure $BMIMBF_4$ was measured as 0.3 mV/K. The origin of the Seebeck coefficient in $BMIMBF_4$ can be attributed to the mobility of cations and anions which produces a steady state of the potential difference between hot and cold electrodes. The value of the ionic Seebeck coefficients of IL and IGs were evaluated from the slope of the plots between open circuit voltage and the corresponding temperature gradient across the device. Good linearity was observed and the slope of all the graphs demonstrated constant values of ionic Seebeck coefficient. Fig. S6 displays a decreasing trend in the ionic Seebeck coefficient with an increase in IL concentration. The ionic Seebeck coefficient of IG 60 was 2.35 mV/K.

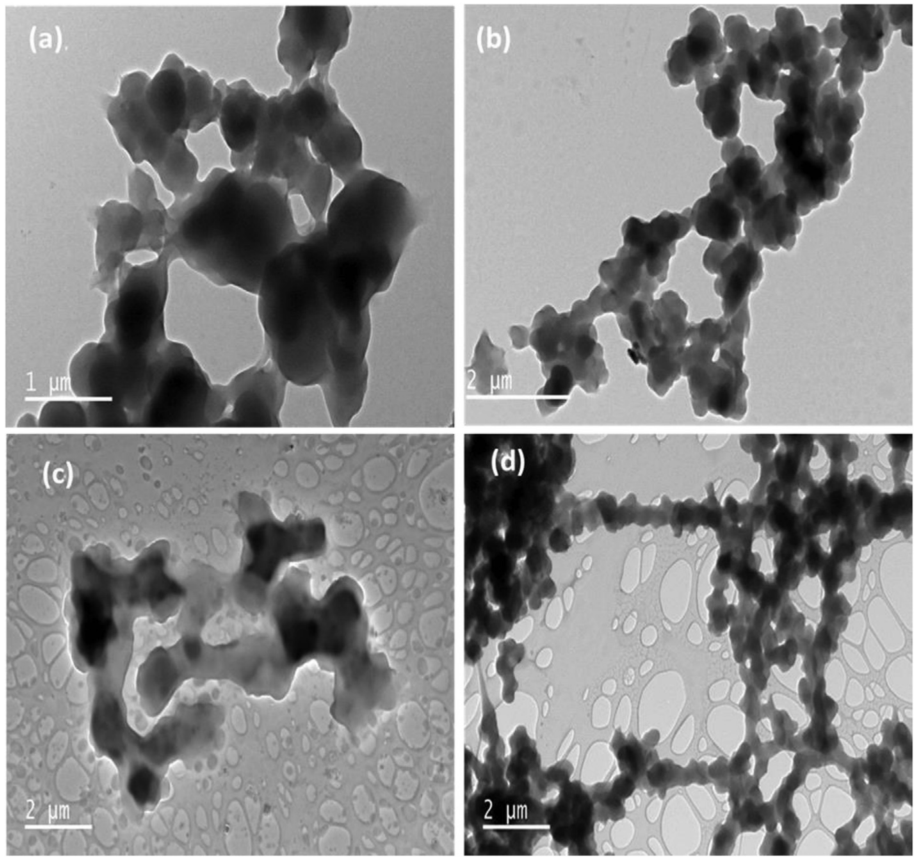

Fig. 8. FETEM images of (a) IG 60, (b) IG 70, (c) IG 80, and (d) IG 90. The microstructure of the IGs was observed by field emission transmission electron microscopy (FETEM, JEM-2100F, JEOL Inc., Japan) at an accelerating voltage of 200 kV.

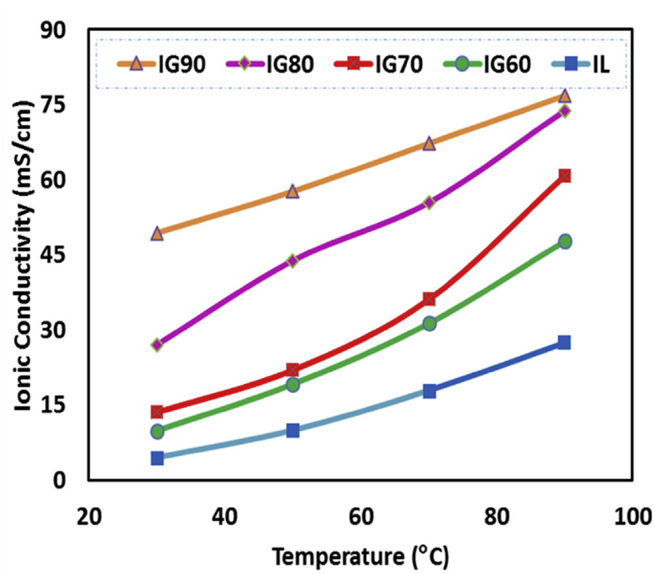

Fig. 9. Temperature dependence of ionic conductivity of IL ($BMIMBF_4$) and four types of Ionogels. In order to find the temperature dependence of impedance of IL and IGs, the coin cells were mounted on a sample holder and heated in an Espec SU-241 temperature chamber. The value of impedance was obtained from Nyquist plots and ionic conductivity was calculated using the Equation (3).

However, the ionic Seebeck coefficient of IG 60 is enhanced seven times as compared with that of pristine $BMIMBF_4$. The reason for the higher value of the Seebeck coefficient of IG 60 as compared with pristine $BMIMBF_4$ can be explained as follows. Wu et al. synthesized the Argyrodite Li6PS5Cl Solid-State Electrolyte and reported a strong correlation of homogeneous distribution and mobility of Li-ions [47]. The vital role of charge carrier mobility in the improvement of the Seebeck coefficient of inorganic and organic conducting thermoelectric materials have been reported in the literature [48,49]. This pathway is basically unexplored for ionogels. As we have observed in the images of EDX mapping of ionogels that cations and anions of the ionic liquid are homogeneously distributed in the PEGDMA matrix. We assume that this homogeneous distribution of cations and anions will increase the mobility of the ions which may be held responsible for higher Seebeck coefficient of ionogels.

With the increase in the concentration of IL the Seebeck coefficient of IGs decreased. The Seebeck coefficient of IG 90 was 0.86 mV/K. As the concentration of $BMIMBF_4$ increases, the number of ions increases and the ionic Seebeck coefficient decreases because of the general tradeoff between electrical conductivity and the Seebeck coefficient of thermoelectric materials. In ionogels similar relation between the ionic conductivity and Seebeck coefficient is observed as in inorganic semiconductors and conducting polymers, both the Seebeck coefficient and electrical conductivity are antagonistically related to each other with respect to carrier concentration [50]. A plot of Seebeck coefficients of IL and four ionogels with varying IL concentration from 60 to 90 wt % is shown in Fig. 10. All the ionic Seebeck coefficients were positive and we termed the ionogel as p-type ionogel. The ionic conductivity of an ionic liquid is mainly correlated to the motion of their cations

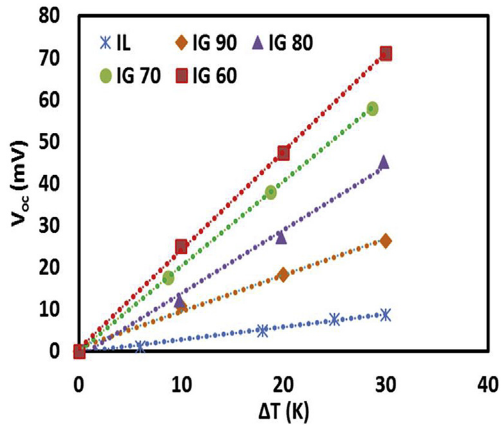

Fig. 10. The Seebeck coefficient of IL and four types of ionogels. The Seebeck coefficient was determined from the slope of the plot between ΔT (V/K) and V_{oc} (mV).

because it has a higher diffusion coefficient than that of the anion. Thus, we can conclude that the positive value of the Seebeck coefficient is due to the motion of the imidazolium cations as majority charge carriers in the ionogels. As it is confirmed from FTIR results that BF_4^- anion makes strong complexes with the polymer matrix and its mobility is less than that of the cation.

3.7. Thermal conductivity

The variation in thermal conductivity of $BMIMBF_4$ and ionogels with temperature has been plotted in Fig. 11. The thermal conductivity of $BMIMBF_4$ at 30 °C was measured as 0.194 W/m.K. The $BMIMBF_4$ demonstrated a very weak temperature dependence of thermal conductivity which is in accordance with the results published by Valkenburg et al. [51]. A slight increase in the values of thermal conductivities of all the ionogels with an increase in temperature has been observed. The thermal conductivities of IG 60, IG 70 and IG 80 were less than that of the ionic liquid. According to Equation (1) in order to achieve a high figure of merit, lower thermal conductivity is desirable. IG 90 exhibited the thermal conductivity values close to that of pure IL. This may be attributed to the very high concentration of IL that the IG behavior for heat transport is similar to that of pure IL. The final value of thermal conductivity was the average of five measurements that were very close to each other with a standard deviation of 0.003 W/m. K which indicates high accuracy of the results measured with THB 500.

3.8. Cyclic voltammetry

The cyclic voltammetry of ionogels in the voltage range from −1.5−1.5 V at a scan rate of 75 mV/s is shown in Fig. 12. With the increase in the concentration of $BMIMBF_4$ the number of ions increase which in turn increase the current flow through IGs. The value of maximum current for IG 60 is 33.4 μA and it increased to 80 μA for the IG 90. The cyclic voltammetry results provide experimental evidence for an increase in ionic conductivity of ionogels with an increase in weight % of $BMIMBF_4$. Fig. S7 shows the effect of scan rate on the current passing through ionogels. With an increase in scan rate the amount of current increases in all IGs because faster scan rate decreases the size of the diffusion layer and higher currents are observed [52].

3.9. Power output

The power output of ionogels was calculated by measuring the potential difference values and known resistance values using the formula, $P = V^2/R$ and is divided by the electrode area for power density calculations. The output power density showed the parabolic relationship as displayed in Fig. 12 and the maximum value of power density achieved was (measured at ΔT = 40 °C) 107 μ W/m^2 for IG 90. The reason behind the high power of IG 90 is that it has more current due to more number of charge carriers as compared with other IGs. This argument is also supported by cyclic voltammetry results of IGs which demonstrated the highest current of 80 μA for IG with highest weight % of $BMIMBF_4$. The thermoelectric cells followed Ohm's law and behave like a simple system, $V = IR$ which implies V is proportional to I and resistance is basically due to charge transport and Ohmic resistances [53]. The power output of IGs is shown in Fig. 13.

3.10. Power factor and figure of merit

The power factor of a thermoelectric material can be calculated by the Equation (5)

$$P = \sigma S^2 \tag{5}$$

The power factor for IG 60 was evaluated as 54.7 mW/m^{-1} K^{-1}. For thermoelectric materials, there is a general tradeoff between

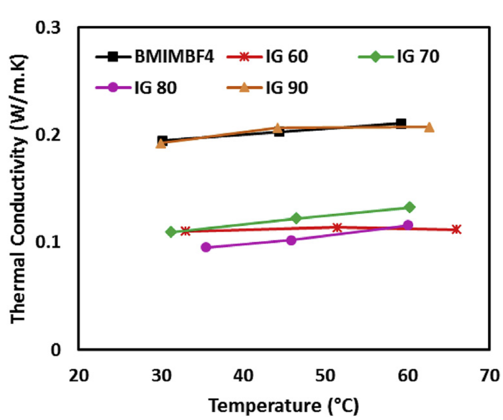

Fig. 11. Temperature-dependent thermal conductivity of $BMIMBF_4$, and four types of ionogels. The Transient Hot Bridge (THB) 500 instrument from Linseis (Germany) with a power heater of 20 mW and measuring the current of 5 mA was employed to measure the thermal conductivities of the IL and ionogels.

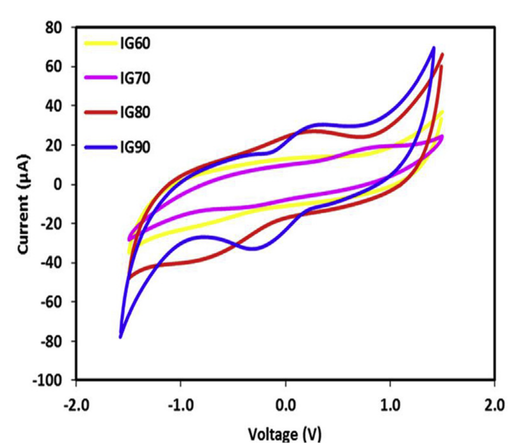

Fig. 12. A comparison of cyclic voltammograms of four types of ionogels at a scan rate of 75 mV/s with a step size of 10 and applied AC voltage of 10 mV.

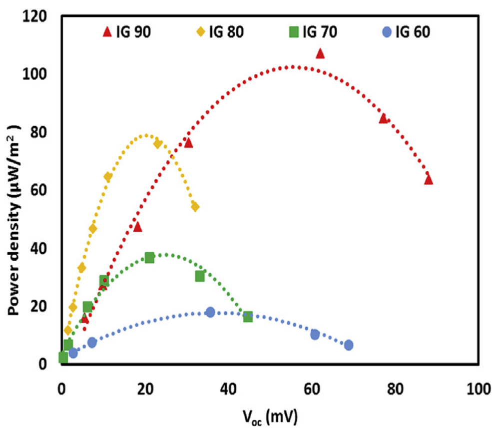

Fig. 13. The power output of the four types of ionogels. The fabricated coin cells were maintained at a temperature gradient of 40 K and connected to an external load. The resulting voltage was measured by Agilent 34461A$6_{1/2}$ digital Multi-meter and power was calculated using the formula, $P=V^2/R$.

electrical conductivity and Seebeck coefficient. The highest Seebeck coefficient of 2.35 mV/K was observed for IG 60 with the lowest ionic conductivity amongst the four ionogels. With the increase in wt. % of IL, the ionic conductivity increased and Seebeck coefficient decreased. The highest power factor of 61.6 $mW/m^{-1}\ K^{-1}$ was witnessed for IG 80. The thermoelectric figure of merit (ZT) for the ionogels at 30 °C was calculated by Equation (1). The highest figure of merit (1.9×10^{-2}) was exhibited by IG 80. This value of ZT is 20 times more than the figure of merit (0.094×10^{-2}) for liquid thermoelectrochemical (TEC) cells prepared by Siddique et al. [23]. Consequently, the ionogels has high thermoelectric performance over liquid TEC along with an advantage of solution of leakage problem due to the solid state of ionogels. The results of the power factor and thermoelectric figure of merit indicate that the 80 wt % of IL is the optimized concentration for the high thermoelectric performance of ionogels.

4. Conclusions

The effects of variation in $BMIMBF_4$ content on the thermoelectric and thermophysical properties of the novel class of ionogels have been successfully investigated. A hostile relationship between the Seebeck coefficient and ionic conductivity of IGs was observed as evidenced in literature for traditional thermoelectric materials. Remarkably, high values of Seebeck coefficient and ionic conductivity of IGs were achieved. Experimental evidence of an increase in ionic conductivity with an increase in the concentration of ionic liquid was observed from decrees in the glass transition temperature of the ionogels. Ionogels were thermally stable up to 390 °C which indicate their exceptionally higher thermal stability. FESEM and FETEM images suggest that ionic liquid is well confined in the PEGDMA scaffold and cross-linking is also evident. Cross-linking of the polymer chain was further confirmed by the quantitative elemental analysis of the links observed in FETEM images by energy dispersive X-ray spectroscopy (EDS). The maximum power density of 107 $\mu W/m^2$ was observed for ionogel with IG 90. The chemically cross-linked ionogels with promising thermoelectric performance may serve as a cheaper and more scalable route for direct conversion of waste heat into electricity. Also, this work can be helpful to have a better understanding of the thermoelectric properties of ionogels. Furthermore, promising thermoelectric properties of ionogels can be explored by changing the type of ionic liquids and varying the polymerization method.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.electacta.2019.134575.

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